UNIVERSITY of HOUSTON | ECE

CEA 3337: Digital Signals & Systems,
Class Worksheet 3

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4 points for each part. 2 points for trying and 2 for full correct answer.

1. (24 points) Plot the following waveforms:

$$x_1(t) = r(t-1) + r(t-2) + r(t-3)$$
$$x_2(t) = r(t) - r(t-2) + r(t-3)$$
$$x_3(t) = r(t) - r(t-1) - r(t-2)$$
$$x_4(t) = rect\left(\frac{t+1}{4}\right)$$
$$x_5(t) = \delta(3t-3)$$
$$x_6(t) = r(t)u(t-1)$$

8 points for each part. 4 points for trying and 4 for full correct answer.

(32 points) Using the properties of the impulse $\delta(t)$, calculate the following integrals:

$$I_1 = 2\int_{-2}^{+2} \delta(t)dt$$
$$I_2 = \int_{-1/2}^{+1/2} \delta(t-2)dt$$
$$I_3 = \int_{-1}^{+1} \delta(3t)dt$$
$$I_4 = \int_{-1}^{+1} \delta(3t-1)dt$$

6 points for each part. 3 points for trying and 3 for full correct answer.

3. (24 points) Plot the following waveforms:

$$x_7(t) = 1/e^{-t}$$
$$x_8(t) = e^{t}u(t)$$
$$x_9(t) = (1+e^{-t})u(t)$$

$$x_{10}(t) = (1+e^{-t+1})u(t-1)$$

10 points for trying and 10 for full correct answer.

4. (20 points) Calculate the average power of the following complex signal. Hint: Use the complex conjugate of the signal.

$$x_{11}(t) = 4(1-j)e^{-j2t}$$